MILITARY STANDARD

TRUCKLOADING OF HAZARDOUS MATERIALS

MIL-STD-1320-121 (NAVY)

29 JANUARY 1974

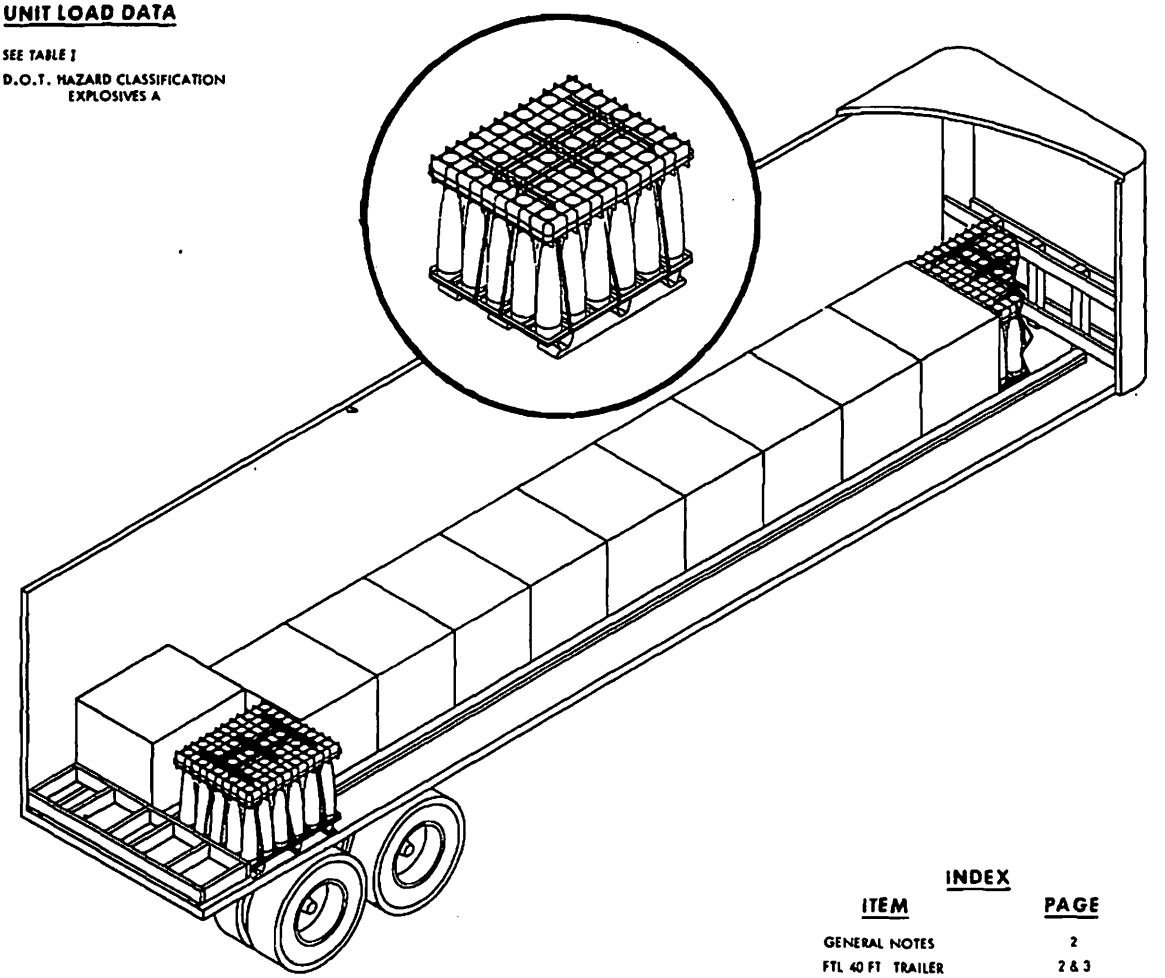
SUPERSEDING WR-51/121 18 JUNE 1970

PROJECTILE, 6" AAC, HC OR AP
IN PALLET ADAPTER MK 20 MOD 0
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

SEE TABLE I

D.O.T. HAZARD CLASSIFICATION
EXPLOSIVES A



INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL 40 FT TRAILER	2 & 3
LTL	4 & 5

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINER-LOADING AND CARLOADING MILITARY STANDARDS. REFER TO *INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).*

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M.S. Gray NWHL 1/28/74
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 1/28/74
SIGNATURE (ORD) SEASCOM, BY DIRECTION DATE

ORIGINATOR Lawrence R. Quiroz 1/28/74
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:44Z
Check the source to verify that this is the current version before use.

1. THE DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING FLEET ISSUE UNIT LOADS OF 6 INCH AAC, HC OR AP PROJECTILES. THE AAC AND HC PROJECTILES SHALL BE PALLETIZED IN ACCORDANCE WITH WR-54/8 AND THE AP PROJECTILES IN ACCORDANCE WITH WR-54/118.
2. PROCEDURES AND PRACTICES ARE INTENDED FOR 40 FT VAN TYPE TRAILERS WITH TANDEM AXLES LOCATED IN THE "WESTERN" POSITION (AT EXTREME REAR OF TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. LENGTH OF STRUTS, ITEM 2, SHALL BE SUCH AS TO CAUSE THE REAR BLOCKING ASSEMBLY TO BEAR AGAINST UNIT LOADS AND THE TRAILER DOORS WHEN THE DOORS ARE IN THE CLOSED POSITION.
4. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
5. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS AND ATTACH "EXPLOSIVES A" PLACARD TO BOTH SIDES, FRONT, AND BACK OF TRAILER.
6. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS FF-N-105, TYPES II, STYLE 10, COMMON BRIGHT.
7. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL 40 FT TRAILER (VAN TYPE)

1. THE FIRST TEN (10) UNIT LOADS ARE POSITIONED ON THE CENTER LINE OF THE TRAILER WITH THE 48 INCH LENGTHS OF THE UNIT LOAD ACROSS THE TRAILER.
2. THE LAST TWO (2) UNIT LOADS ARE POSITIONED TWO ACROSS THE TRAILER WITH THE 48 INCH LENGTH OF THE UNIT LOAD LONGITUDINAL IN THE TRAILER.

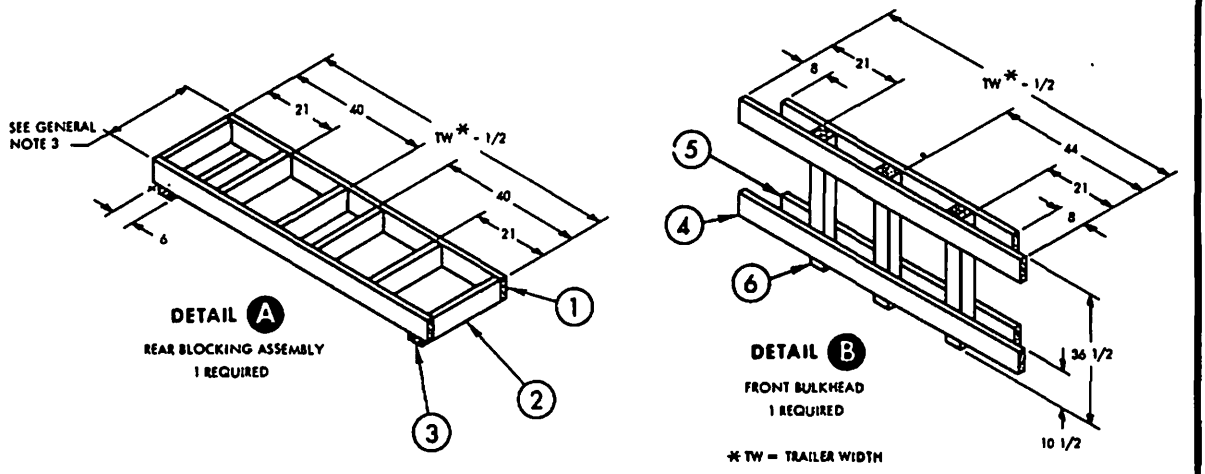
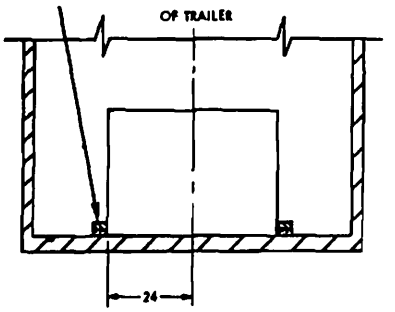


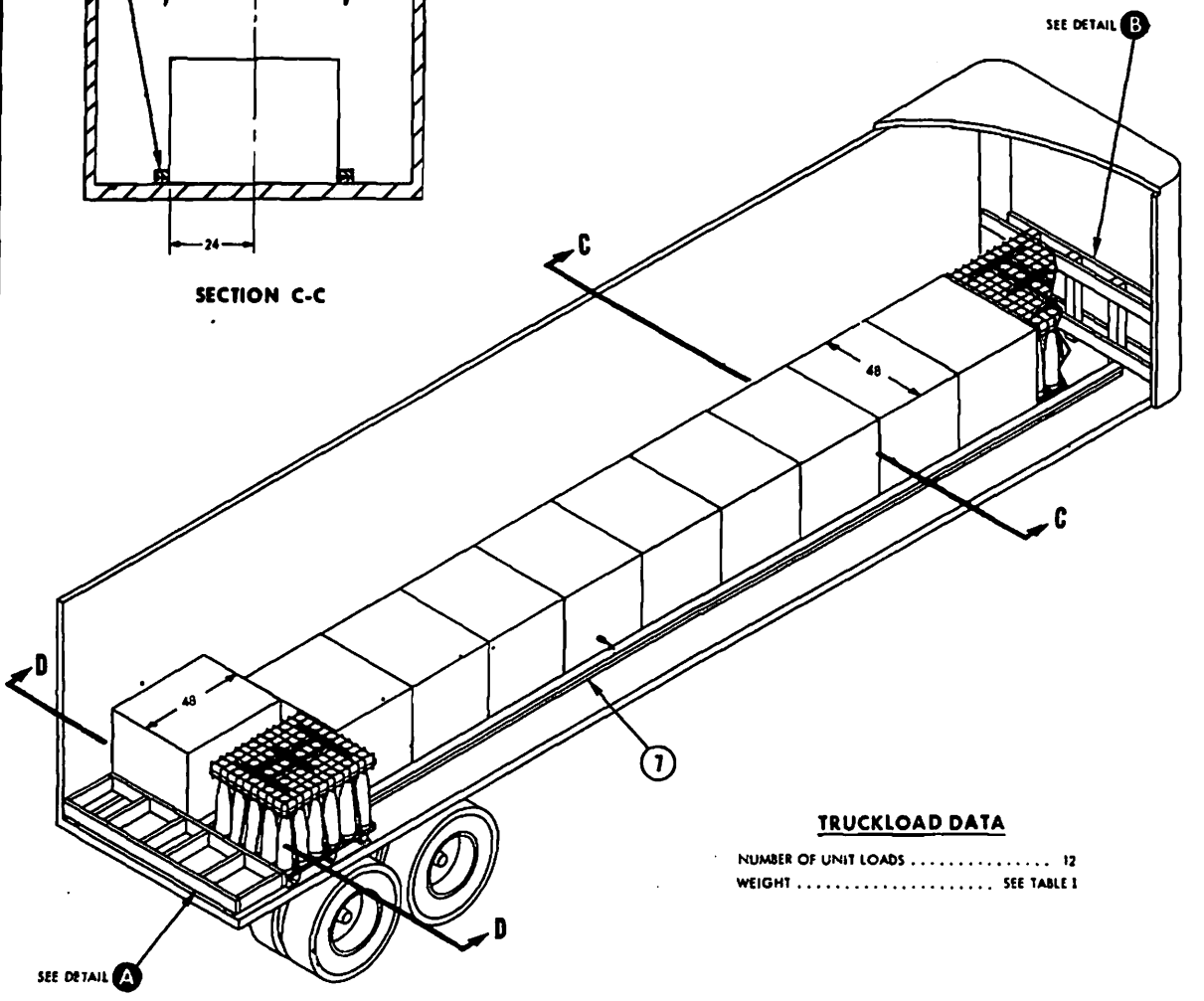
TABLE I

ITEM	UNIT LOAD DWG. NO.	UNIT LOAD DIMENSIONS	UNIT LOAD CUBE (CU.FT.)	UNIT LOAD WT. (LBS.)	TRUCKLOAD WT. (LBS.)
AAC OR HC PROJ. W/CAP MK 1 OR 3	WR-54/8	48 L x 40 W x 35 H	38.8	3440	41, 280
AAC OR HC PROJ. W/CAP MK 4	WR-54/8	48 L x 40 W x 37 5/8 H	41.8	3440	41, 280
AP PROJECTILE	WR-54/118	48 L x 40 W x 32 3/4 H	36.4	3450	41, 400

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:44Z
Check the source to verify that this is the current version before use.

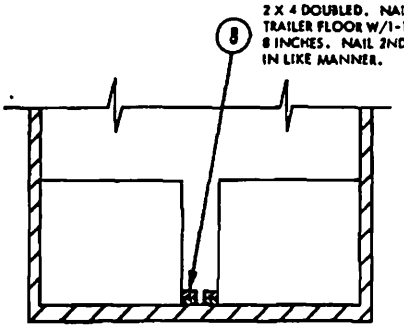


SECTION C-C



TRUCKLOAD DATA

NUMBER OF UNIT LOADS 12
WEIGHT ..... SEE TABLE I



SECTION D-D

* TW = TRAILER WIDTH

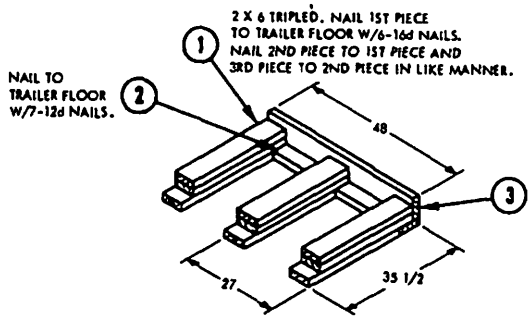
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
8	SLEEPER	2 X 4 X 48	4	SEE SECTION D-D		
7	SLEEPER	2 X 4 X RANDOM LENGTHS	135 FT	SEE SECTION C-C		
6	UPRIGHT	4 X 4 X 36 1/2	3	SEE 4 & 5		
5	FWD CROSS MEMBER	2 X 6 TO SUIT	2	6	3/JOINT	12d
4	AFT CROSS MEMBER	2 X 6 X TW*-1/2	2	6	3/JOINT	12d
3	LEG	2 X 4 TO SUIT	2	1	2/JOINT	12d
2	STRUT	2 X 6 TO SUIT	6	SEE 1	-	-
1	CROSS MEMBER	2 X 6 X TW*-1/2	2	2	3/JOINT	16d

LIST OF MATERIAL AND NAILING DATA

FTL 40 FT TRAILER (VAN TYPE)

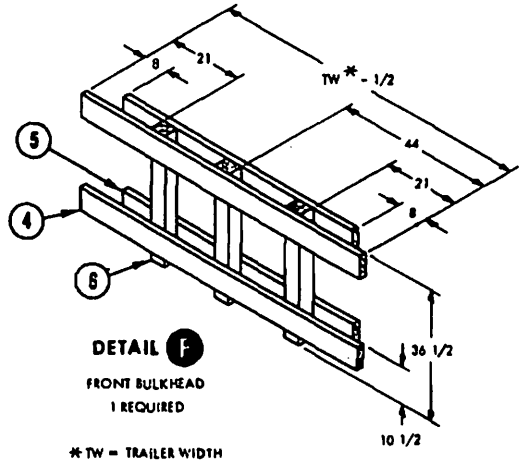
Source: https://assist.dla.mil -- Downloaded: 202640-23T23:44Z
Check the source to verify that this is the current version before use.

1. WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD, ALL UNIT LOADS ARE POSITIONED ON THE CENTER LINE OF THE TRAILER AS SHOWN IN SECTION G-G. THE 48 INCH DIMENSION OF ALL UNIT LOADS ARE POSITIONED ACROSS THE TRAILER. BLOCK AT REAR OF LOAD USING REAR BLOCKING ASSEMBLY, DETAIL E.
2. THE LTL SHOWN ON PAGE 5 IS FOR SIX UNIT LOADS. THIS DOES NOT LIMIT LTL SHIPMENTS TO THIS QUANTITY. THEY MAY CONSIST OF A GREATER OR LESSER NUMBER OF UNIT LOADS.
3. ACTUAL SHIPPING REQUIREMENTS WILL DETERMINE THE NUMBER OF UNIT LOADS TO BE SHIPPED. THE LOAD PATTERN AND DUNNAGING SHALL BE IN ACCORDANCE WITH THE APPROPRIATE PARTS OF THIS DOCUMENT AND THE GENERAL TRUCKLOADING STANDARD, MIL-STD-1320 (NAVY).



DETAIL E

REAR BLOCKING ASSEMBLY
1 REQUIRED



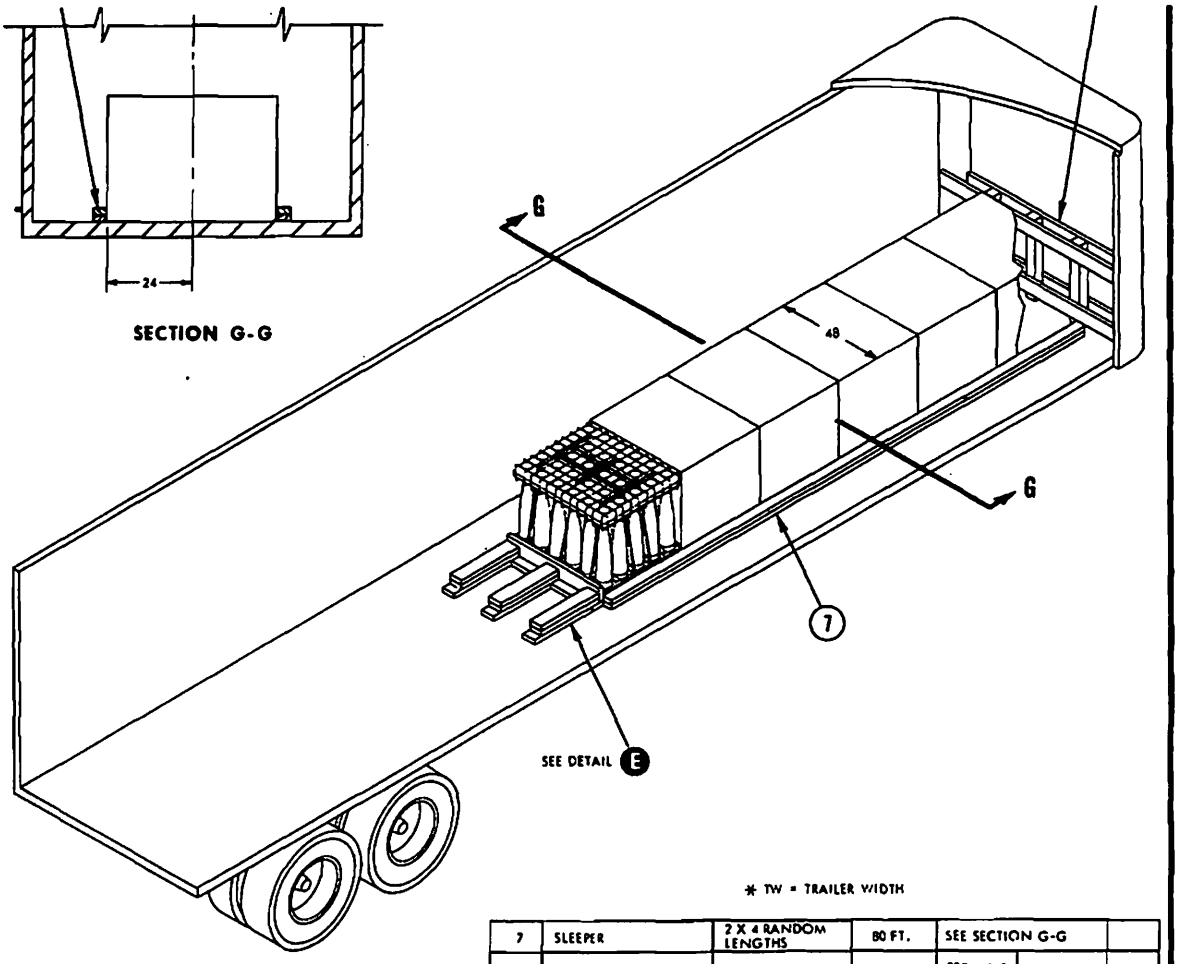
DETAIL F

FRONT BULKHEAD
1 REQUIRED

* TW = TRAILER WIDTH

Source: https://assist.dla.mil -- Downloaded: 202640-23T23:44Z
Check the source to verify that this is the current version before use.

SECTION G-G



* TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
LIST OF MATERIAL AND NAILING DATA						
7	SLEEPER	2 X 4 RANDOM LENGTHS	80 FT.	SEE SECTION G-G		
6	VERTICAL	4 X 4 X 36 1/2	3	SEE 4 & 5	-	-
5	FWD CROSS MEMBER	2 X 6 TO SUIT	2	6	3/JOINT	12d
4	AFT CROSS MEMBER	2 X 6 X TW* -1/2	2	6	3/JOINT	12d
3	UPPER CROSS MEMBER	2 X 6 X 48	1	SEE 2	-	-
2	LOWER CROSS MEMBER	2 X 6 X 48	1	3	6	16d
1	BACKUP CLEAT	2 X 6 X 30	9	SEE DETAIL E		

LTL ALL LENGTH TRAILERS

REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N224)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:44Z
Check the source to verify that this is the current version before use.